

ODDSCAPE PLACE REPORTS

Reading the Landscape

OPR-001

Ring of Brodgar

The Ceremonial Landscape within the Ness of Brodgar

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Terrain Interpretation • Landscape Visualisation • Place Analysis

Foreword

Every landscape tells two stories.

The first is immediately visible. We see hills, water, vegetation, buildings and monuments arranged across the land as they exist today. This is the landscape encountered by walkers, visitors and photographers; the landscape experienced from eye level.

The second story is less obvious. It lies beneath the vegetation, beneath modern roads, beneath centuries of agricultural change. It is written in the shape of the land itself. Subtle ridges direct movement. Shallow depressions collect water. Slight changes in elevation alter visibility across surprisingly large distances. These relationships are often too gradual to appreciate from the ground yet become strikingly clear when viewed through a Digital Terrain Model.

Oddscape Place Reports are built upon the belief that understanding a landscape begins with understanding its terrain.

The Ring of Brodgar has been studied for generations as one of Europe's most remarkable prehistoric monuments. Archaeologists have investigated its construction, chronology and cultural significance. Historians have explored its place within the wider Neolithic landscape of Orkney. Their work forms the foundation upon which our understanding of Brodgar rests.

This publication asks a different question.

Rather than beginning with the stones, it begins with the land.

Why does this place exist here?

Why did generations of people continue to invest meaning in this particular landscape?

What qualities of terrain, water, visibility and movement distinguish this basin from countless other places across Britain?

These are questions of landscape rather than archaeology. They do not replace established scholarship, nor do they seek to reinterpret evidence beyond its limits. Instead, they offer another perspective—one that starts with the physical structure of the land and examines how that structure may have influenced the human experience of place.

The purpose of this report is therefore not simply to describe the Ring of Brodgar.

Its purpose is to read the landscape.

Understanding the Place

To many visitors, the Ring of Brodgar appears almost unexpectedly modest.

Unlike Stonehenge, it does not dominate an open plain through sheer monumentality. Unlike many hillforts, it occupies no commanding summit. It is neither concealed within woodland nor dramatically framed by mountains. Instead, it rests quietly upon a narrow strip of gently undulating land between two broad lochs.

At first glance, this seems an unlikely location for one of Britain's greatest ceremonial monuments.

Yet this apparent simplicity is deceptive.

The surrounding landscape possesses an extraordinary degree of internal coherence. The Ring occupies a natural corridor between the Loch of Harray to the north and the Loch of Stenness to the south. These two water bodies almost meet, leaving only a narrow land bridge through which movement naturally converges. Roads still follow this alignment today, demonstrating that the logic of the terrain has changed remarkably little over thousands of years.

From ground level, these relationships are easily overlooked. Attention is naturally drawn towards the standing stones themselves. Visitors move through the monument, photograph individual stones and admire the surrounding views, but rarely perceive the wider structure that binds the entire landscape together.

Digital terrain modelling changes that perspective.

When modern visual distractions are removed, the basin reveals itself as a carefully organised composition of water, low relief and long horizons. The Ring becomes one element within a much larger environmental system rather than an isolated archaeological feature.

This shift in perspective is central to the Oddscape approach.

Rather than asking how the monument relates to its immediate surroundings, we ask how the surrounding landscape gives meaning to the monument.

Reading the Landscape

One of the greatest strengths of LiDAR-derived terrain modelling is its ability to reveal subtle topographic relationships that are almost impossible to appreciate from eye level.

The landscape surrounding the Ring of Brodgar is often described as relatively flat. Technically this is true. Elevation changes are modest, particularly when compared with Britain's mountainous regions. Yet "flat" is not the same as featureless.

Across the ceremonial basin, differences in height of only a few metres create surprisingly powerful effects.

Gentle rises provide dry routes through wetter ground.

Shallow depressions influence seasonal drainage.

Low ridges shape views across the lochs.

Slight terraces define movement without the need for constructed boundaries.

These are not dramatic landforms. Their importance lies precisely in their subtlety.

Landscape architects often speak of the difference between relief and character. A landscape need not possess dramatic relief to possess a powerful identity. In many cases, the most memorable places are those where delicate variations in terrain organise experience almost unconsciously.

Brodgar exemplifies this principle.

Viewed through the terrain model, the monument occupies a position that balances openness with enclosure. The surrounding land neither overwhelms nor disappears. Instead, it creates a continuous visual dialogue between water, horizon and stone.

The Digital Terrain Model therefore becomes more than a technical representation of elevation.

It becomes a means of understanding why this landscape feels the way it does.

Landscape Observation 01

The Ring of Brodgar does not dominate its landscape through height.

Instead, it occupies one of the most visually connected positions within the ceremonial basin.

Its significance arises not from commanding the landscape, but from participating within it.

1. Geological Foundations

Every landscape begins long before the arrival of people.

Long before the first standing stone was erected, before ceremonial processions crossed the isthmus, and before the first fields were enclosed, the landscape of Brodgar had already undergone hundreds of millions of years of change. The basin in which the Ring

stands is not accidental; it is the product of geological processes that established the physical framework within which later human activity unfolded.

The bedrock beneath the Heart of Neolithic Orkney belongs principally to the Devonian Old Red Sandstone, deposited approximately 385 million years ago when this part of Scotland lay close to the equator. Vast freshwater lakes occupied the Orcadian Basin, accumulating alternating layers of fine muds and sands that would eventually become the flagstones and sandstones characteristic of Orkney today.

These rocks possess qualities that have profoundly influenced both the appearance of the landscape and the history of its occupation. Their relatively horizontal bedding creates broad, gently undulating landforms rather than steep mountains or deeply incised valleys. They weather into fertile soils while also providing abundant building stone that can be split into remarkably regular slabs. The very geology that shaped the landscape also supplied the material from which its monuments were constructed.

Yet the landscape visible today owes as much to ice as it does to rock.

Repeated glaciations during the Quaternary Period transformed the subdued Devonian terrain into the basin now occupied by the Lochs of Harray and Stenness. Ice sheets moving across Orkney exploited weaknesses within the bedrock, deepening natural depressions while smoothing surrounding ridges. As the glaciers retreated, meltwater further modified the landscape, leaving behind the subtle topography that characterises the area today.

Unlike alpine environments, where glaciers produce dramatic peaks and valleys, Orkney's glacial legacy is restrained. Relief is modest, gradients are gentle and transitions are gradual. This apparent simplicity is deceptive. It creates precisely the sort of landscape in which very small differences in elevation assume great importance.

The Ring of Brodgar occupies one such subtle threshold.

Geological Time and Human Time

One of the remarkable aspects of Brodgar is the contrast between geological and human timescales.

The rocks beneath the monument record processes measured in hundreds of millions of years. The shaping of the basin by glacial ice occurred over tens of thousands of years. Human occupation, by comparison, occupies only the most recent moments of this immense chronology.

Yet these vastly different timescales are inseparable.

Without Devonian sedimentation there would be no flagstones.

Without glaciation there would be no lochs.

Without the lochs there would be no narrow isthmus.

Without the isthmus it is difficult to imagine the ceremonial landscape developing in quite the same way.

Landscape architecture often concerns itself with designing places that endure beyond individual lifetimes. Brodgar reminds us that every designed landscape, no matter how ancient, exists within a much longer geological narrative that ultimately determines what is possible.

Reading Bedrock Through Terrain

Digital Terrain Models do not show geology directly.

Instead, they reveal geology through its influence upon landform.

Across the Brodgar basin the terrain displays broad convex ridges separated by shallow depressions that gently collect surface water. These forms are not random. They reflect the interaction between resistant bedrock, glacial erosion and subsequent weathering.

The resulting terrain possesses an unusual quality.

Rather than forcing movement through narrow valleys or across dramatic passes, it encourages gradual transitions. Routes emerge naturally from slight differences in elevation. Visibility changes almost imperceptibly as one walks across the landscape. The experience is one of continuous unfolding rather than sudden revelation.

Such landscapes reward careful observation.

What appears unremarkable at first glance gradually reveals increasing complexity the longer one studies it.

This characteristic lies at the heart of the Oddscape approach.

Terrain modelling allows subtle landscapes to speak with greater clarity.

Rather than exaggerating relief, it enables the viewer to appreciate relationships that already exist but are easily overlooked. It transforms gentle undulations into legible forms, helping us understand why particular places became significant long before the written record began.

Landscape Observation 02

The geology of Brodgar does not merely provide the setting for the monument; it creates the conditions that made the ceremonial landscape possible.

The Ring of Brodgar should therefore be understood not simply as an archaeological site, but as one expression of a much older geological landscape whose structure continues to influence movement, visibility and the experience of place.

2. Water as Landscape Structure

If geology provides the foundations of a landscape, water gives it life.

Throughout history, rivers, estuaries, lochs and coastlines have directed settlement, movement and trade. They have provided food, defined territorial boundaries and shaped patterns of communication. At the Ring of Brodgar, however, water performs an additional role. It becomes an architectural element within the landscape itself.

The Ring stands between two very different bodies of water.

To the north lies the Loch of Harray, Scotland's largest freshwater loch by area. Its broad, open surface creates a sense of expansive calm, reflecting the constantly changing skies of Orkney and extending the apparent width of the landscape.

To the south lies the Loch of Stenness. Although often described simply as a loch, it occupies a transitional position between freshwater and the sea. Tidal influence, combined with shallower waters and changing salinity, creates a more dynamic environment whose ecological character differs noticeably from its northern neighbour.

Between these contrasting water bodies lies the narrow strip of land upon which the Ring of Brodgar stands.

It is this relationship—not simply the monument itself—that defines the landscape.

The Landscape Narrows

When viewed through a terrain model, the geography of Brodgar becomes immediately apparent.

The two lochs almost meet.

Only a comparatively narrow corridor of slightly higher ground separates them, forming a natural isthmus across which movement is gently concentrated. Unlike mountain passes or river crossings that announce themselves dramatically, this narrowing is understated. Its influence emerges gradually through experience rather than spectacle.

Modern roads continue to occupy this corridor.

Visitors approaching the Ring today unknowingly follow routes dictated by the same physical constraints that have shaped movement for thousands of years.

This persistence illustrates an important principle within landscape analysis.

While human infrastructure changes repeatedly through history, the fundamental logic of terrain remains remarkably constant. Roads, tracks and pathways often evolve, yet they continue to respond to the same slopes, water margins and dry ground that influenced earlier generations.

The land remembers.

Water as Enclosure

One of the most striking observations emerging from terrain analysis is that the Ring occupies an unusually balanced position between openness and containment.

At first glance, the surrounding landscape appears almost limitless. Long horizons extend across the lochs towards distant hills, producing an exceptional sense of openness.

Yet psychologically the space feels surprisingly enclosed.

This apparent contradiction arises because water acts as a natural edge.

Unlike woodland or steep hills, water does not obstruct vision. Instead, it defines space without closing it. The lochs frame the ceremonial basin in much the same way that the walls of a great room define an interior without necessarily dominating it.

From within the Ring, the observer experiences a landscape that is simultaneously expansive and contained.

Few locations achieve this balance so effectively.

Reflection and Perception

Modern visitors often underestimate the influence of water because they experience it only as scenery.

For prehistoric communities, its presence would have been far more immediate.

The lochs reflected light across the basin throughout the day.

Wind continually altered their appearance.

Mist drifted across the water during changing weather.

Seasonal variations affected shorelines and vegetation.

The landscape was never static.

Instead, it existed in a continual state of subtle transformation.

The standing stones themselves would have appeared differently according to season, weather and time of day, their visual presence constantly modified by reflections, cloud shadows and changing atmospheric conditions.

This dynamic quality remains one of the defining characteristics of Brodgar today.

Terrain analysis reminds us that landscapes are not fixed compositions.

They are living systems whose appearance changes continuously while their underlying structure remains remarkably stable.

Water and Ceremony

It is impossible to know precisely how Neolithic communities understood these waters.

Speculation should never replace evidence.

Nevertheless, landscape analysis allows certain observations to be made with confidence.

The Ring occupies neither the centre of a dry plain nor the edge of a single water body.

Instead, it exists within a landscape organised around relationships between land and water.

Movement towards the monument inevitably involves passing between the lochs.

Views across the basin are continually framed by water.

The monument's position cannot be separated from these relationships because they define the character of the place itself.

Whether or not symbolic meanings were attached to these waters lies beyond the scope of terrain analysis.

What can be stated confidently is that water was not incidental to the experience of Brodgar.

It was fundamental.

Terrain Observation 03

Water at Brodgar should not be interpreted merely as landscape setting. It functions as landscape structure.

The Lochs of Harray and Stenness organise movement, frame views, define spatial enclosure and contribute directly to the enduring character of the ceremonial basin. Their influence is inseparable from that of the monument itself.

Looking Beyond the Monument

Perhaps the greatest lesson offered by the terrain is this:

The Ring of Brodgar is not the landscape.

It is one carefully positioned element within a much larger environmental composition.

The land, the water, the horizon and the routes through the basin all work together to create an experience of place that no single feature could achieve alone.

Seen from this perspective, the stones become less an isolated destination and more a point of convergence—a place where the wider landscape gathers into focus.

3. Movement Through the Basin

Every landscape encourages certain journeys while discouraging others.

In mountainous regions, routes are often obvious. Valleys funnel movement. Ridges provide natural corridors. Rivers dictate crossing points. The landscape announces where people should walk.

The ceremonial basin surrounding the Ring of Brodgar is different.

Its guidance is almost invisible.

Nothing within the terrain compels movement dramatically. Instead, the land offers subtle preferences. Slight rises remain drier underfoot. Gentle gradients reduce effort. Water

margins quietly constrain direction without appearing restrictive. These influences are so understated that many visitors remain unaware they are following them.

Yet they are remarkably persistent.

The modern road passing the Ring of Brodgar follows broadly the same corridor dictated by the terrain for thousands of years. Although its precise alignment reflects engineering decisions made in recent centuries, its overall position owes far more to geography than to design.

The land had already made many of the important decisions.

Walking the Landscape

Terrain models possess an advantage unavailable to conventional maps.

Rather than representing movement as coloured lines drawn across paper, they allow movement to be imagined as an embodied experience.

One begins to ask different questions.

Where would the ground remain driest after prolonged rain?

Where would a gentle rise provide a wider field of view?

Where would livestock naturally choose to cross?

Where would carrying heavy materials become easiest?

These questions shift attention away from monuments and towards everyday interaction with the landscape.

The Ring of Brodgar was not approached through empty space.

Every journey towards it unfolded through terrain already shaping perception long before the monument came into view.

The gradual emergence of the stones across open ground would have formed part of that experience.

Unlike monuments hidden within woodland or revealed suddenly over hilltops, Brodgar announces itself progressively.

Distance becomes part of the architecture.

Corridors of Movement

Viewed from above, the basin reveals several subtle corridors where movement naturally converges.

These corridors are not roads in the modern sense.

They are areas where combinations of elevation, drainage and visibility make passage consistently attractive.

The Ring occupies one of the most significant of these intersections.

Its location therefore offers two complementary advantages.

First, it is easily reached.

Second, it is encountered naturally.

This distinction matters.

Some monuments require deliberate effort to visit.

Others lie along journeys people already wish to make.

The Ring of Brodgar appears to belong largely to the second category.

Its builders selected a place already embedded within the logic of the landscape.

The monument strengthened an existing relationship rather than creating an entirely new one.

The Experience of Arrival

Modern visitors usually arrive by car.

Parking areas, roads and footpaths compress what would once have been a gradual transition into a journey lasting only a few minutes.

Terrain modelling encourages us to slow that experience mentally.

Imagine approaching on foot.

The lochs remain visible for much of the journey.

The land rises almost imperceptibly.

Individual stones appear one by one against the horizon.

There is no dramatic reveal.

Instead, anticipation grows steadily.

Landscape architects often describe this process as sequential experience.

Places are rarely understood from a single viewpoint.

They are revealed through movement.

The Ring of Brodgar rewards precisely this kind of reading.

Every step alters relationships between stone, water and horizon.

Every slight rise offers a subtly different composition.

The landscape is experienced not as a static picture but as a sequence of unfolding moments.

Movement Beyond Ceremony

Although Brodgar is rightly recognised for its ceremonial significance, the surrounding landscape would also have supported ordinary activities.

People travelled between settlements.

Livestock required grazing.

Resources needed transporting.

Water had to be accessed.

Children played.

Communities gathered.

Ceremonial landscapes do not exist separately from everyday life.

Instead, they become woven into its rhythms.

One of the strengths of terrain interpretation is that it reminds us of this continuity.

The same dry ridge serving daily travel may also become the route towards a monument.

The same open basin supporting grazing may also provide an appropriate setting for communal gatherings.

The landscape does not divide itself neatly into sacred and ordinary.

Human activity creates those distinctions.

The terrain simply provides opportunities.

Time Written Into Routes

Perhaps the most remarkable observation is the endurance of movement itself.

Over five thousand years, societies have changed beyond recognition.

Languages have disappeared.

Belief systems have transformed.

Technologies have advanced.

Yet people continue to cross the basin in remarkably similar ways.

This persistence illustrates one of landscape's most enduring qualities.

While cultures evolve, terrain changes slowly.

It continues quietly influencing movement long after the reasons for particular journeys have been forgotten.

Routes become traditions.

Traditions become roads.

Roads become assumptions.

Eventually, the influence of the land becomes almost invisible precisely because it has proved so successful.

The Ring of Brodgar occupies a landscape already predisposed to movement.

Rather than imposing order upon an empty landscape, the monument appears to strengthen relationships that already existed between water, dry ground and the natural corridors connecting them.

Its significance arises not only from where it stands, but from how people arrive.

Reading Movement Through Terrain

The Digital Terrain Model reminds us that movement is rarely arbitrary.

Every path represents countless small decisions made in response to slope, water, visibility and effort.

At Brodgar those decisions accumulated over thousands of years, creating a landscape whose routes feel almost inevitable.

The monument therefore occupies more than a geographical position.

It occupies a position within movement itself.

To understand the Ring fully is not merely to stand within it.

It is to walk towards it, to experience the gradual unfolding of the basin, and to recognise how subtly the terrain has been guiding that journey all along.

Landscape Reflection

There is a temptation when studying prehistoric monuments to think in terms of destinations.

Terrain encourages a different perspective.

The journey is part of the monument.

Every step across the basin prepares the visitor for arrival.

Every slight rise changes perception.

Every glimpse of water alters spatial awareness.

By the time the Ring is reached, the landscape has already begun telling its story.

The stones simply complete the sentence.

4. Visibility & Intervisibility

One of the greatest misconceptions about prehistoric monuments is that their significance can be understood by studying each site in isolation.

The Ring of Brodgar demonstrates precisely the opposite.

When viewed through the terrain rather than through the monument alone, it becomes clear that Brodgar forms part of a much larger spatial composition. The Standing Stones of Stenness, Maeshowe, the Ness of Brodgar and numerous lesser-known archaeological

sites occupy positions that are neither random nor independent. They exist within the same landscape framework, connected by visibility, movement and the subtle geometry of the basin itself.

Terrain modelling allows these relationships to be appreciated at a scale that is difficult to achieve from ground level.

Rather than seeing isolated monuments, we begin to recognise a connected ceremonial landscape.

Seeing Without Dominating

It would be easy to assume that important ceremonial sites should occupy the highest available ground.

The Ring of Brodgar challenges this assumption.

The Digital Terrain Model shows that the monument does not stand upon a summit. Higher land exists nearby. If elevation alone had been the objective, alternative locations would have provided greater dominance over the surrounding countryside.

Instead, Brodgar occupies something more subtle.

Its position provides extensive visual connection while avoiding excessive prominence. The monument is visible without becoming isolated. It participates in the landscape rather than overwhelming it.

This distinction is important.

A commanding hilltop often separates itself from everyday movement.

Brodgar remains part of it.

Its builders appear to have chosen a place that balanced accessibility with visibility, openness with enclosure and prominence with integration.

This balance may be one of the defining characteristics of the entire ceremonial basin.

The Basin as a Theatre

Landscape architects sometimes describe enclosed landscapes as outdoor rooms.

The Brodgar basin suggests something slightly different.

It functions more like a natural theatre.

The lochs define its edges.

Low ridges shape the audience's perspective.

The horizon provides an uninterrupted backdrop.

Within this setting the Ring occupies neither centre stage nor the highest balcony. Instead, it sits at precisely the point where the surrounding landscape achieves its greatest coherence.

This observation does not imply deliberate theatrical intent by Neolithic builders.

Rather, it recognises that certain landscapes possess an inherent spatial order that later human activity naturally adopts and reinforces.

The Ring appears to strengthen an already powerful composition rather than inventing one.

Long Horizons

One of Orkney's defining qualities is the remarkable openness of its horizons.

Trees are relatively scarce.

Relief is gentle.

The eye travels great distances across land and water.

At Brodgar these long views become part of the experience of place.

The horizon is never static.

Clouds continually reshape it.

Changing weather alters its colour and apparent distance.

Light reflected from the lochs creates subtle shifts throughout the day.

Unlike mountainous landscapes, where attention is often directed upwards, Brodgar encourages lateral vision.

The landscape unfolds horizontally.

This produces an unusual psychological effect.

Rather than feeling confined within a valley or elevated above surrounding terrain, the observer experiences an almost continuous visual dialogue between land, water and sky.

The standing stones punctuate that dialogue rather than interrupting it.

Intervisibility

The concept of intervisibility refers to the ability of different locations to remain visually connected.

Within the Heart of Neolithic Orkney this relationship has long attracted archaeological interest.

Terrain analysis provides an additional perspective.

Rather than asking simply whether two monuments can see one another, it asks how the landscape mediates that relationship.

Do low ridges partially conceal certain approaches?

Do shallow rises reveal monuments gradually rather than immediately?

How does water influence visual continuity?

These questions shift attention from binary visibility towards experiential visibility.

The Digital Terrain Model suggests that the basin encourages a sequence of carefully balanced visual connections.

Monuments appear, disappear and re-emerge as movement continues through the landscape.

The terrain itself becomes part of the choreography.

Reading Space

One of the most striking qualities revealed by terrain modelling is the generosity of space.

Modern aerial photography often compresses landscapes.

Distances appear shorter.

Relationships appear more immediate.

Walking the terrain produces a different understanding.

The basin is spacious without becoming empty.

Its openness allows monuments to maintain distinct identities while remaining visually related.

No single feature overwhelms the others.

Instead, each contributes to a wider network whose coherence emerges only when viewed as a whole.

This is perhaps the greatest lesson offered by Brodgar.

The landscape is not simply a collection of important places.

It is itself the important place.

Landscape Character Through Visibility

Landscape character is created not only by physical landforms but by how those landforms shape perception.

Visibility is therefore a fundamental component of character.

At Brodgar, visibility is characterised by:

- **exceptionally long horizontal views;**
- uninterrupted relationships between land and water;
- **gradual visual revelation through movement;**
- **subtle enclosure despite apparent openness;**

- strong connections between neighbouring ceremonial sites.

These characteristics combine to produce a landscape whose identity is remarkably consistent despite its apparent simplicity.

The terrain possesses clarity without monotony.

Every slight change in position produces new relationships while preserving the overall coherence of the basin.

Terrain Observation 05

The Ring of Brodgar is not visually dominant because it is high. It is visually significant because it occupies one of the most connected positions within the ceremonial landscape.

Its power derives from relationship rather than elevation.

The monument belongs to a network of places held together by the structure of the terrain itself.

The Landscape as Memory

Perhaps this explains why Brodgar continues to resonate.

The monument is unforgettable.

Yet the memory people carry away is often larger than the circle itself.

They remember the light upon the lochs.

The broad sky.

The wind.

The immense openness.

The slow approach across gently rising ground.

The standing stones become inseparable from the landscape that surrounds them.

Terrain modelling simply makes those relationships visible.

It allows us to recognise what visitors have been feeling for generations without necessarily understanding why.

The landscape has always been doing the work.

The monument merely gives it a centre.

5. Landscape Character Assessment

Landscape character is more than scenery.

It is the distinct and recognisable pattern created by the interaction of geology, topography, water, vegetation, land use and cultural history. While archaeological monuments often become the focus of attention, they form only one component of a much broader landscape character.

The Ring of Brodgar occupies a landscape whose identity extends far beyond the circle itself.

The ceremonial basin possesses a remarkable consistency of form. Water, low relief and expansive horizons combine to create an environment that feels both open and intimate, ancient yet continually changing. It is a landscape that reveals its complexity gradually, rewarding patience rather than demanding attention.

From a landscape architectural perspective, Brodgar demonstrates how powerful subtle landscapes can become when their individual components work in harmony.

Scale

Scale is one of the defining qualities of the Brodgar basin.

Unlike mountainous regions where relief dominates perception, Brodgar achieves its sense of grandeur through breadth rather than height.

The eye is encouraged to travel across the landscape rather than upwards.

The lochs stretch horizontally.

The surrounding farmland extends uninterrupted towards distant hills.

The sky occupies almost half the visual composition.

This produces a feeling of spaciousness that is disproportionate to the relatively modest physical dimensions of the basin.

Large landscapes are not always created by large landforms.

Sometimes they are created by uninterrupted continuity.

Enclosure

At first glance, Brodgar appears almost entirely open.

Yet this openness conceals an equally important quality.

The basin possesses a subtle sense of enclosure.

Water defines its boundaries.

Low ridges gently contain views without restricting them.

Distant hills provide a soft visual edge rather than an abrupt barrier.

This creates what might be described as psychological enclosure.

The observer feels part of a coherent landscape while remaining constantly aware of the wider environment beyond.

This balance is rare.

Too much enclosure limits experience.

Too little creates visual ambiguity.

Brodgar occupies an unusually successful middle ground.

Pattern

Every landscape possesses pattern.

At Brodgar those patterns emerge through repetition rather than contrast.

Broad water bodies alternate with gently rising land.

Pasture follows subtle changes in topography.

Stone walls reinforce natural field boundaries.

Archaeological monuments occupy positions that appear visually related without becoming mechanically ordered.

Nothing feels arbitrary.

Equally, nothing feels excessively formal.

The landscape possesses a quiet rhythm that becomes increasingly apparent through repeated observation.

This rhythm is one of its greatest strengths.

Tranquillity

Modern assessments of landscape character frequently include tranquillity as an indicator of environmental quality.

Although influenced today by roads, visitors and agricultural activity, Brodgar retains a remarkable sense of calm.

Several factors contribute to this experience.

The broad openness of the basin reduces feelings of confinement.

Water moderates visual complexity.

The absence of intensive urban development preserves exceptionally dark night skies.

Wind, birds and changing weather provide the dominant sounds rather than continuous mechanical noise.

These characteristics combine to create a landscape whose atmosphere remains remarkably consistent with its physical form.

Tranquillity here is not silence.

It is continuity.

Legibility

Landscape architects often describe certain landscapes as being highly legible.

A legible landscape is one that can be understood intuitively.

Visitors instinctively recognise where they are, how movement occurs and how different elements relate to one another.

The Ring of Brodgar demonstrates exceptional legibility.

The lochs provide orientation.

The gentle relief explains movement.

The horizon remains continuously visible.

Even without maps, the landscape is relatively easy to interpret.

This clarity contributes significantly to its enduring appeal.

People rarely become disoriented.

Instead, they develop an immediate understanding of the basin's overall structure.

Sense of Place

Perhaps no concept is more central to Oddscape than sense of place.

This report has deliberately avoided presenting Brodgar as merely an archaeological site because its significance cannot be reduced to individual stones or historical events.

Sense of place emerges from relationships.

The interaction between water and land.

The sequence of arrival.

The changing qualities of light.

The continuity of movement.

The openness of the horizon.

The presence of monuments within a landscape already possessing remarkable coherence.

Each element reinforces the others.

Remove one and the character changes.

Together they create a place that continues to resonate across generations.

This understanding lies at the heart of landscape architecture.

Places are experienced as wholes.

Not collections of isolated features.

Reading Character Through Terrain

Terrain modelling contributes something unique to landscape character assessment.

Traditional surveys rely heavily upon field observation.

LiDAR allows those observations to be tested against the underlying physical structure of the land.

The subtle ridges directing movement become visible.

Drainage patterns become legible.

Relationships between monuments and topography emerge with unusual clarity.

The terrain does not replace observation.

It strengthens it.

The result is a richer understanding of why this landscape feels coherent despite its apparent simplicity.

Landscape Character Summary

Landscape Type: Open Ceremonial Basin

Key Characteristics

- Broad horizontal horizons.
- Gently undulating glacial landforms.
- Freshwater and tidal lochs defining landscape structure.
- Long-distance visibility.
- Low woodland cover.
- Strong relationship between archaeology and topography.
- High landscape legibility.
- Exceptional sense of place.
- Quiet but powerful spatial composition.

Terrain Observation 06

The defining quality of Brodgar is not monumentality. It is coherence.

Every component of the landscape—geology, water, relief, movement and visibility—works together to produce a place that feels remarkably complete.

The Ring of Brodgar succeeds because it belongs to that landscape rather than simply occupying it.

A Landscape Designed by Time

One of the enduring lessons of Brodgar is that the greatest landscapes are rarely created in a single moment.

They emerge through successive layers of natural process and human response.

Geology established the foundations.

Ice shaped the basin.

Water defined movement.

People recognised the qualities already present.

The monument became another layer within a landscape whose composition had begun millions of years earlier.

Seen through this perspective, the Ring of Brodgar is not simply an ancient monument.

It is part of a conversation between people and landscape that continues today.

Every visitor who walks between the lochs becomes part of that same story.

6. Terrain Interpretation

Reading What the Land Reveals

There is a temptation when working with Digital Terrain Models to treat them as simply another form of mapping.

Contours become smoother.

Hillshades become more attractive.

Three-dimensional models become easier to visualise.

Yet if terrain modelling is used only to produce better images, much of its potential remains unrealised.

The purpose of terrain interpretation is different.

It asks not simply "What does the land look like?"

Instead it asks:

"What is the land telling us?"

This distinction lies at the heart of the Oddscape approach.

The Ring of Brodgar provides an ideal example because its significance cannot be explained through elevation, geology or archaeology alone. Each discipline contributes an important part of the story, but none fully explains why this particular landscape possesses such enduring coherence.

Terrain interpretation attempts to bridge those perspectives.

Rather than examining each component separately, it considers how they interact.

Removing the Noise

One of the first experiences many people have when viewing a LiDAR-derived terrain model is surprise.

Familiar landscapes suddenly appear unfamiliar.

Modern roads disappear.

Buildings lose their dominance.

Field boundaries become less important.

Vegetation no longer obscures subtle landforms.

The landscape begins to resemble something older.

Not prehistoric.

Simply more fundamental.

It becomes easier to distinguish those features created by natural process from those introduced by successive generations of human activity.

This does not diminish archaeology.

On the contrary, it allows archaeology to be understood within its physical setting.

The monument returns to its landscape.

The Basin Emerges

Perhaps the most significant observation arising from the Ring of Brodgar terrain model is the clarity with which the ceremonial basin becomes visible.

Walking through the landscape, this basin is difficult to perceive.

Human vision operates close to ground level.

Our experience is shaped by hedgerows, roads, walls, vegetation and the constantly changing direction of travel.

The terrain model removes these distractions.

Suddenly the relationship between the Loch of Harray, the Loch of Stenness and the intervening land becomes unmistakable.

The Ring no longer appears isolated.

It occupies the narrowest and most visually connected part of a coherent environmental system.

This observation is not speculative.

It arises directly from the geometry of the terrain.

The basin already existed.

The monument simply occupies one of its most significant positions.

A Landscape of Small Differences

Mountain landscapes impress through dramatic relief.

The Brodgar basin demonstrates a different kind of complexity.

Here, differences measured in only one or two metres influence movement across kilometres.

A slight rise remains dry after prolonged rainfall.

A shallow hollow channels seasonal drainage.

A gentle terrace alters the apparent height of the horizon.

Such changes are almost impossible to appreciate from conventional photography.

Terrain modelling magnifies their importance without exaggerating their scale.

This is perhaps one of the greatest strengths of LiDAR.

It teaches us to notice subtlety.

Reading Movement Without Roads

One of the unexpected advantages of terrain modelling is the ability to imagine movement before modern infrastructure.

The Digital Terrain Model contains no tarmac.

No painted lines.

No road signs.

Yet likely routes begin to reveal themselves naturally.

The landscape quietly suggests where people would choose to walk.

This does not mean every prehistoric path can be reconstructed.

Such certainty would be impossible.

Instead, the terrain reveals probability.

Dry ground becomes attractive.

Steep slopes become inconvenient.

Visibility influences comfort and orientation.

Movement becomes understandable through the physical qualities of the landscape itself.

Modern roads merely confirm many of these observations.

Water as Architecture

Throughout this publication, water has repeatedly emerged as the organising element of the Brodgar landscape.

Terrain interpretation strengthens that conclusion.

The Ring does not simply overlook water.

Neither does it merely occupy land between two lochs.

Instead, the lochs define the geometry of the entire basin.

Their margins determine where movement becomes easiest.

Their surfaces influence visibility.

Their presence creates enclosure without obstruction.

They perform precisely the role that walls, courtyards and boundaries perform within designed architecture.

The landscape possesses structure.

Water provides much of that structure.

Landscape Before Monument

Perhaps the most important conclusion arising from terrain interpretation is that the landscape possesses meaning independently of the Ring itself.

Imagine removing every standing stone.

The basin would remain.

The lochs would still frame movement.

The ridges would still organise routes.

The horizons would continue to define space.

The place would retain an unusual degree of coherence.

Now imagine the opposite.

Place the Ring upon an ordinary agricultural field elsewhere in Britain.

Much of its landscape significance would disappear.

This thought experiment illustrates a central principle.

The monument derives part of its meaning from the landscape.

The landscape does not derive all of its meaning from the monument.

The relationship works in both directions.

Reading Relationships

Traditional site analysis often concentrates upon individual features.

Terrain interpretation concentrates upon relationships.

Stone relates to water.

Water relates to movement.

Movement relates to visibility.

Visibility relates to topography.

Topography relates to geology.

No single relationship explains the landscape.

Together they create something greater than the sum of their parts.

This systems-based perspective is increasingly important within contemporary landscape practice.

Complex landscapes rarely depend upon one dominant factor.

They emerge through interaction.

The Ring of Brodgar demonstrates this principle with remarkable clarity.

What the Terrain Cannot Tell Us

Terrain interpretation has limits.

The Digital Terrain Model cannot explain belief.

It cannot reveal ritual.

It cannot determine symbolic meaning.

It cannot tell us why particular ceremonies occurred or what stories prehistoric communities associated with the landscape.

These questions remain the domain of archaeology, anthropology and cultural interpretation.

Recognising these limits is essential.

Oddscape does not seek to replace specialist disciplines.

Instead, it contributes another layer of understanding.

Terrain explains possibility.

Human culture explains purpose.

The two should always be read together.

Terrain Observation 07

The greatest contribution of terrain modelling is not that it produces beautiful images.

It reveals relationships that are difficult to perceive from ground level, allowing landscapes to be understood as connected systems rather than collections of isolated features.

At Brodgar, those relationships consistently point towards one conclusion:

The monument belongs exactly where it stands.

Not because it occupies the highest ground.

Not because it dominates the landscape.

But because it occupies the point at which geology, water, movement and visibility achieve their greatest balance.

The Landscape Speaks First

Long before the first stone was raised, the basin already possessed structure.

People did not create that structure.

They recognised it.

That recognition may be one of humanity's oldest design skills.

The Ring of Brodgar reminds us that successful places are often those that begin by listening to the land rather than attempting to overwrite it.

Perhaps this is the deepest lesson offered by terrain interpretation.

The landscape is not simply the stage upon which history unfolds.

It is one of the principal authors of the story itself.

The Oddscape Perspective

The Landscape is the Monument

There is a tendency when visiting places such as the Ring of Brodgar to assume that the monument is the destination and the landscape merely the route taken to reach it.

It is an understandable assumption. The standing stones are visually striking, historically significant and internationally recognised. They appear in guidebooks, photographs and documentaries. They have become the symbol by which the landscape itself is remembered.

Yet after spending time with the terrain, another possibility begins to emerge.

Perhaps the monument has never been the whole story.

Perhaps the landscape itself is the true monument.

Standing within the Ring, it is easy to become absorbed by the stones.

Standing above the Ring—through the perspective offered by terrain modelling—the emphasis shifts.

The eye moves naturally towards the surrounding basin.

The lochs begin to define the space.

The ridges quietly organise movement.

The horizon becomes part of the composition.

The monument remains important, but it no longer stands alone.

Instead, it becomes one carefully positioned element within a landscape that already possessed extraordinary qualities.

This change in perspective is subtle, yet profound.

It asks us to value not only what people built, but also what they recognised.

Recognition Rather Than Imposition

Modern design often begins with intervention.

We identify a site.

We decide its purpose.

We reshape the land.

We construct.

The Ring of Brodgar suggests another approach.

Its builders may first have recognised an exceptional place before deciding how best to respond to it.

Rather than imposing significance upon an ordinary landscape, they appear to have acknowledged significance that already existed.

The monument strengthened an existing relationship between land, water and movement.

It did not create that relationship from nothing.

This idea resonates strongly within contemporary landscape practice.

The most successful places rarely fight against the landscape.

They emerge through understanding it.

The terrain becomes collaborator rather than obstacle.

Reading Instead of Looking

Most visitors look at landscapes.

Few are encouraged to read them.

Looking is immediate.

Reading requires patience.

It asks us to notice subtle relationships rather than dramatic features.

To recognise patterns repeated across different scales.

To understand that what appears insignificant in isolation may become profoundly meaningful when connected with everything around it.

Terrain modelling supports this slower way of seeing.

It strips away distraction without removing complexity.

Instead of overwhelming the viewer with information, it reveals structure.

The landscape becomes legible.

Not simplified.

Simply easier to understand.

Time Within the Landscape

One of the most remarkable qualities of Brodgar is its relationship with time.

Geological time shaped the basin.

Ecological time continues to shape the lochs.

Human time introduced monuments, paths and settlements.

Visitor time unfolds over hours.

Each operates simultaneously.

The landscape does not privilege one timescale over another.

Instead, it contains them all.

This layered understanding transforms how we experience place.

The Ring ceases to be simply five thousand years old.

It becomes the latest visible chapter within a landscape whose history stretches back hundreds of millions of years.

Every visitor briefly joins that continuing story.

Why This Place Still Matters

Many prehistoric sites survive because they are protected.

The Ring of Brodgar survives for another reason.

It continues to make sense.

Its relationship with the surrounding landscape remains remarkably intact.

The lochs still define movement.

The horizons still organise perception.

The terrain still reveals the same quiet logic that first attracted people thousands of years ago.

This continuity allows modern visitors to experience something surprisingly close to the essential structure encountered by earlier generations.

That continuity is increasingly rare.

It deserves recognition.

What Oddscape Contributes

Oddscape does not seek to replace archaeology.

Nor geology.

Nor history.

Each discipline already explains Brodgar with considerable depth and authority.

Oddscape contributes something different.

It asks the landscape itself to speak.

Digital Terrain Models allow us to observe relationships that are often hidden beneath vegetation, modern infrastructure and the limitations of ground-level experience.

The objective is not to discover new facts for the sake of novelty.

It is to deepen appreciation for places we thought we already understood.

Sometimes understanding grows not from finding something new, but from seeing something familiar differently.

A Different Way of Seeing

Every generation develops new tools for observing landscapes.

Theodolites transformed surveying.

Aerial photography transformed mapping.

Satellite imagery transformed environmental monitoring.

LiDAR has transformed our ability to understand terrain.

Yet technology alone is never enough.

The important question is not what the tool can produce.

It is what the tool allows us to notice.

The value of terrain modelling lies not in the three-dimensional models themselves.

Its value lies in encouraging better observation.

Technology becomes worthwhile when it helps us become more attentive to the landscapes we already inhabit.

The Future of Reading Landscapes

The Ring of Brodgar is only the beginning.

Across Britain there are countless landscapes whose stories remain partially hidden within their terrain.

River valleys.

Estuaries.

Ancient woodland.

Industrial landscapes.

Mountain passes.

Coastlines.

Urban landforms.

Each possesses relationships waiting to be recognised.

The purpose of the Oddscape Place Reports is not simply to document these places.

It is to encourage a different habit of seeing.

To ask questions before drawing conclusions.

To understand landscape as process rather than scenery.

To recognise that every place possesses a structure extending far beyond its most famous landmark.

Closing Reflection

There is a moment that often occurs when studying a terrain model.

Without warning, the landscape begins to make sense.

What previously appeared random acquires order.

Movement becomes understandable.

Water explains itself.

The position of a monument suddenly feels inevitable.

Nothing has changed.

The land has not moved.

No new evidence has appeared.

Only perception has shifted.

Perhaps that is the true purpose of landscape interpretation.

Not to change the landscape.

But to change the way we see it.

Final Observation

The Ring of Brodgar is not remarkable because it contains a stone circle.

It is remarkable because it demonstrates how people recognised the extraordinary qualities of an already extraordinary landscape.

The monument remains one of Britain's great prehistoric achievements.

The landscape remains one of Britain's great masterpieces.

The two cannot be separated.

That is what the terrain reveals.

Some landscapes are visited. Others are understood.

Oddscape exists to help reveal the difference.